

Dong-Kyum Kim

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📍 Max Planck Institute for Security and Privacy, Universitätsstraße 140, 44799 Bochum, Germany

Research Interests

- **Interdisciplinary physicist and AI researcher**, guided by physical principles and observations from neuroscience.
- **Knowledge in large language models**: uncovering how models represent, retrieve, update, and forget information through mechanistic interpretability, model editing, and machine unlearning.
- **From physical and biological systems to AI**: drawing on nonequilibrium statistical physics and biological learning and memory to build safer, more reliable, and controllable AI systems.

Education

Korea Advanced Institute of Science and Technology (KAIST) 2016–2022

Ph.D. in Physics

Advisor: Prof. [Hawoong Jeong](#)

Dissertation: *Nonequilibrium Statistical Physics Study using Deep Learning*

Seoul National University (SNU)

2011–2015

B.S. in Physics, Minor in Computer Science & Engineering

Research Experience

Postdoctoral Researcher, Max Planck Institute for Security and Privacy (MPI-SP) Oct. 2024–Present

- Hosted by Prof. [Meeyoung Cha](#) (Scientific Director)
- Data Science for Humanity Group

Postdoctoral Researcher, Institute for Basic Science (IBS)

Mar. 2022–Sep. 2024

- Hosted by Prof. [Meeyoung Cha](#) (Chief Investigator)
- Data Science Group, Center for Mathematical and Computational Science

Data Science Intern, Samsung Electronics

Sep.–Dec. 2017

- Built distributed ML systems with [Daniel Kim](#) (Senior Data Scientist): multi-GPU anomaly image classification (Keras & Spark) and an Elasticsearch-based image search framework

Publications

[†] denotes equal contribution.

- [1] Jea Kwon, **Dong-Kyum Kim**, Jiwon Kim, Yonghyun Kim, Woong Kook, and Meeyoung Cha. “[AI Engram: In Search of Memory Traces in Artificial Intelligence](#).” *ICML 2026 (Oral)*.
- [2] Minsung Kim, **Dong-Kyum Kim**, Jea Kwon, Nakyeong Yang, Kyomin Jung, and Meeyoung Cha. “[How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models](#).” *ACL 2026*.
- [3] **Dong-Kyum Kim**, Minsung Kim, Jea Kwon, Nakyeong Yang, and Meeyoung Cha. “[Bilinear representation mitigates reversal curse and enables consistent model editing](#).” *ICLR 2026*.
- [4] Nakyeong Yang, **Dong-Kyum Kim**, Jea Kwon, Minsung Kim, Kyomin Jung, and Meeyoung Cha. “[Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning](#).” *ICLR 2026*.
- [5] Jea Kwon, Jiwon Kim, **Dong-Kyum Kim**, and Meeyoung Cha. “[Moir: Let the Model Direct Its Own Story for Robust Cross-Domain Knowledge Editing](#).” *ICML 2026 Workshop on Mechanistic Interpretability*.
- [6] Minsung Kim, Jea Kwon, **Dong-Kyum Kim**, and Meeyoung Cha. “[In Search of the Engram in LLMs](#).” *ICLR 2025 Blogpost*.
- [7] Gwangsu Kim, **Dong-Kyum Kim**, and Hawoong Jeong. “[Spontaneous emergence of rudimentary music detectors in deep neural networks](#).” *Nature Communications* 2024.

- [8] Jea Kwon, Sunpil Kim, **Dong-Kyum Kim**, Jinhyeong Joo, SoHyung Kim, Meeyoung Cha, and C. Justin Lee. “[SUBTLE: An unsupervised platform with temporal link embedding that maps animal behavior.](#)” *IJCV* (2024).
- [9] **Dong-Kyum Kim**[†], Jea Kwon[†], Meeyoung Cha, and C. Justin Lee. “[Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity.](#)” *NeurIPS* 2023.
- [10] Sangyun Lee, **Dong-Kyum Kim**, Jong-Min Park, Won Kyu Kim, Hyunggyu Park, and Jae Sung Lee. “[Multidimensional entropic bound.](#)” *Phys. Rev. Research* **5**, 013194 (2023).
- [11] **Dong-Kyum Kim**[†], Sangyun Lee[†], and Hawoong Jeong. “[Estimating entropy production with odd-parity state variables via ML.](#)” *Phys. Rev. Research* **4**, 023051 (2022).
- [12] Youngkyoung Bae, **Dong-Kyum Kim**, and Hawoong Jeong. “[Inferring dissipation maps from videos using CNNs.](#)” *Phys. Rev. Research* **4**, 033094 (2022).
- [13] **Dong-Kyum Kim** and Hawoong Jeong. “[Deep reinforcement learning for feedback control in a collective flashing ratchet.](#)” *Phys. Rev. Research* **3**, L022002 (2021).
- [14] **Dong-Kyum Kim**[†], Youngkyoung Bae[†], Sangyun Lee, and Hawoong Jeong. “[Learning Entropy Production via Neural Networks.](#)” *Phys. Rev. Lett.* **125**, 140604 (2020).
- [15] **Dong-Kyum Kim**[†], Byunghwee Lee[†], Daniel Kim, and Hawoong Jeong. “[Multi-label classification of historical documents using hierarchical attention.](#)” *J. Korean Phys. Soc.* **76**, 368 (2020).

Technical Skills

- **Daily driver:** Currently running multiple Claude Code sessions (Max plan, \$200/mo) via tmux as my primary research workflow.
- **ML/Interp:** PyTorch, JAX, TransformerLens, & nnsight.
- **Infrastructure:** Distributed training training & inference (Slurm system with hundreds of H100s), large-scale data pipelines

Academic Service

Reviewer: ICML 2026 Workshop Mechanistic Interpretability, ICML 2026 (Received Gold Reviewer Award), ICLR 2026, NeurIPS 2025, ICML 2025 Workshop World Models, ICLR 2025 Workshop Re-Align, NeurIPS 2024 Workshop Behavioral ML, ICML 2024 Workshop LLMs and Cognition.

Awards

Pre-doctoral Fellow of Physics, KAIST

2021

References

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